MAWHIBA GRADE 9 · UNIT 2 LINEAR FUNCTIONS · ACTIVITY 4

Sequences Generated by Repeated Application of a Linear Function

Advanced Supplementary Mathematics
Mawhiba Schools Partnership

Student ..... **Class** ..... **Date** ..... / ..... / 2026 **Teacher** Mr Malek Thiab

UNIT 2 LEARNING OBJECTIVES

- Identify the properties of linear functions and their graphs, including composite and inverse functions
- Use geometrical and algebraic reasoning to express generalisations and prove results

ADVANCED PERFORMANCE CHARACTERISTICS

- Reason with algebra and make links with geometrical representations
- Appreciate the importance of mathematical explanation and proof

You will need A spreadsheet or graph plotter of your choice, a calculator, a ruler and a sharp pencil for the grid on page 2.

TASK 1 · PRACTICE · START AT 1 AND KEEP GOING

Starting with $x = 1$ *, investigate the sequence of numbers generated by repeated application of the function* $f(x) = \frac{x}{2} + 3$ *.*

You might choose to use a spreadsheet to help you calculate the terms of the sequence: $1,\ f(1),\ ff(1),\ fff(1),\ \ldots$

What happens if you start with other values of x *, for example,* $x = 2$ *or* $x = 9$ *?*

Column 0 is the start value. Column 1 is the value after the function has been applied once, and so on. Write exact decimals.

START	0	1	2	3	4	5	6	7	8
$x = 1$									
$x = 2$									
$x = 9$									
$x = ?$									

What I notice. Where do the terms seem to be heading from each start? From which side do they approach? Does any start behave differently?

One more start of my own. Choose a start that is negative or very large. Does the pattern still hold? Write what you found.

My tool. Which tool did I choose and why was it a good choice?

TASK 2 · APPLY

Four Functions, One Grid

TASK 2 · APPLY · WRITE, SIMPLIFY AND PLOT

Write down and simplify expressions for the functions $f(x)$, $ff(x)$, $fff(x)$ *and* $ffff(x)$. *Plot their graphs on one set of axes.*

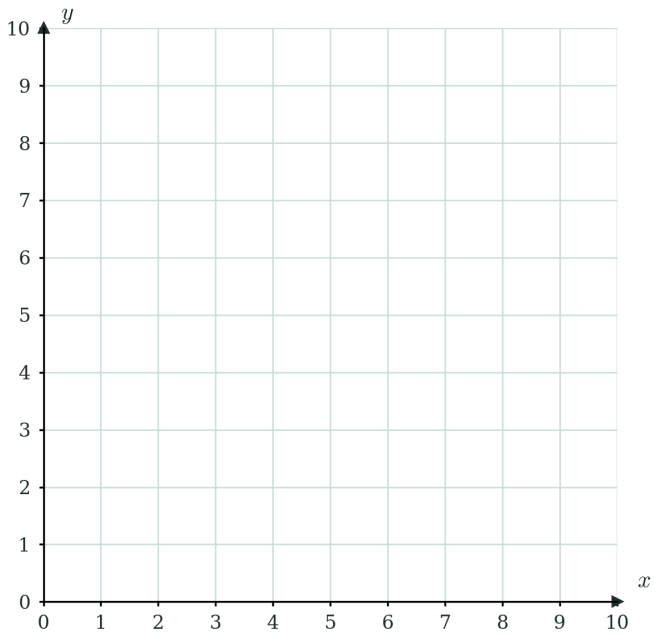
Explain what you notice about these functions and why the solution to $f(x) = x$ *gives the limiting value of the sequence.*

FUNCTION	simplified form
$f(x) =$	
$ff(x) =$	
$fff(x) =$	
$ffff(x) =$	

Work out each one from the one before: apply the function to the previous answer. Check each by substituting a start value from Task 1.

Working space.

Plot all four graphs. Use a ruler, label every line, and mark any point they share.



EXPLAIN · WHAT I NOTICE

1. What do the four graphs have in common, and how do they change from one to the next?

2. Where on your graph can you find the terms of the Task 1 sequence that started at 1? Mark them.

3. Explain why the solution of $f(x) = x$ gives the limiting value of the sequence. Use your graph in your explanation.

TASK 3 · INVESTIGATE

Other Linear Functions

TASK 3 · INVESTIGATE · TRY OTHER FUNCTIONS

In shorthand, we can write $ff(x) = f^2(x)$, $fff(x) = f^3(x)$, $\ldots$ *Investigate the sequence* $x,\ f(x),\ f^2(x),\ f^3(x),\ \ldots$ *for other linear functions. Explain any general results that you find.*

Work in a crew of three or four. Each crew member takes two cards, then you pool your results. Write the value that the function maps onto itself, or write none. Choose your own start value and list six terms (columns 1 to 6).

CARD	function	maps onto itself at	start	1	2	3	4	5	6	what the terms do
A	$f(x) = 2x + 1$									
B	$f(x) = x + 3$									
C	$f(x) = \frac{x}{4} + 3$									
D	$f(x) = \frac{3x}{4} + 2$									
E	$f(x) = -\frac{x}{2} + 6$									
F	$f(x) = -x + 5$									
G	$f(x) = -2x + 3$									
H	$f(x) = 2x - 4$									

Sorting. Group the eight cards into families by what their terms do. What decides the family: the gradient, the intercept, or both?

General results. Write what you have found for any linear function, and test it on a function that is not on the cards.

TASK 3 · INVESTIGATE, THEN APPLY

Algebra and a Nursery

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TASK 3 · INVESTIGATE · EXPLAIN WITH ALGEBRA

Take the general linear function $f(x) = mx + c$ *. Find expressions for the repeated applications and look for the pattern.*

REPEATED APPLICATION	expanded and simplified
$f^2(x) =$	
$f^3(x) =$	
$f^4(x) =$	
$f^n(x) =$	

My explanation. Use your formula to explain what happens to the terms when the gradient is large, small or negative. Test it on two of your cards.

APPLY · A JEDDAH TREE NURSERY

Model values: a Jeddah nursery grows saplings for tree-planting projects in the spirit of the Saudi Green Initiative. Each year 20% of its saplings are lost and 500 new ones are planted. These numbers are illustrative, not real nursery data.

$n =$ saplings now, $\quad f(n) =$ saplings next year

$f(n) = 0.8n + 500$

1 · The nursery starts with 1000 saplings. How many are there after 1, 2 and 3 years?

2 · In the long run, how many saplings will it hold? Show the method from Task 2. Does the answer depend on the start?

3 · What if only 10% were lost each year? Predict the long-run number first, then check by calculating.

MY MAWHIBA RECORD

Keep this with your cumulative record. Tick the value you showed today and write one example.

☐ Inquiry ☐ Risk taking ☐ Creativity ☐ Perseverance ☐ Collaboration ☐ Concern for society

Done when… you have three start values tabulated and can say what the terms get close to, you can explain with your graphs why the value that the function maps onto itself is the limit, you have described the behaviours you found with evidence from at least two cards, and you can find the long-run number in the nursery model.